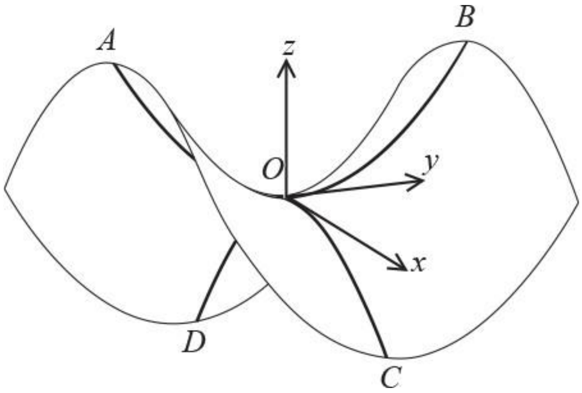
OCR Further Mathematics
Additional Pure Mathematics
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a)



A student wishes to model the saddle of a horse. They use a surface described by a function of the form $z = f(x, y)$ with a saddle point at the origin O. The z-axis is vertically upwards. The x- and y-axes lie in a horizontal plane, with the x-axis across the horse and the y-axis along the length of the horse (see diagram).

The arc AOB is part of a parabola which lies in the yz-plane. The arc COD is part of a parabola which lies in the xz-plane. The saddle is symmetric in both the xz-plane and yz-plane.

The length of the saddle, the distance AB, is to be 0.6 m with both A and B at a height of 0.27 m above O. The width of the saddle, the distance CD, is to be 0.5 m with both C and D at a depth of 0.4 m below O.

On separate diagrams, sketch the sections $x = 0$ and $y = 0$. [2]

(b) Determine a function f that describes the saddle. [You do not need to state the domain of function f.]

[5]

2(a)

For each value of k the sequence of real numbers $\{u_n\}$ is given by $u_1 = 2$ and $u_{n+1} = \dfrac{k}{6+u_n}$. For each of the following cases, either determine a value of k or prove that one does not exist.

$\{u_n\}$ is constant. [2]

(b) $\{u_n\}$ is periodic, with period 2. [3]

(c) $\{u_n\}$ is periodic, with period 4. [5]

3(a) Solve the second-order recurrence system $H_{n+2} = 5H_{n+1} - 4H_n$ with $H_0 = 3$, $H_1 = 7$ for $n \geq 0$. [5]

(b)

(i) Write down the quadratic residues modulo 10.

[1]

(ii) By considering the sequence $\{H_n\}$ modulo 10, prove that H_n is never a perfect square.

[6]

4(a)

Let $I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \, dx$ for integers $n \geq 0$.

Show that, for $n \geq 2$, $nI_n = (n - 1) I_{n-2}$. [4]

(b) Use this reduction formula to deduce the exact value of I_8. [2]

(c)

Use the results of the above parts to determine the exact value of $\int_0^{\frac{1}{2}\pi} \cos^6 x \sin^2 x \, dx$. [2]

5(a) The binary operation ◊ is defined on the set ℂ of complex numbers by

$(a + \mathrm{i}\, b) ◊ (c + \mathrm{i}\, d) = ac + \mathrm{i}(b + ad)$

where a, b, c and d are real numbers.

Is ℂ closed under ◊? Justify your answer. [1]

(b) Prove that ◊ is associative on ℂ. [4]

(c) Determine the identity element of ℂ under ◊. [2]

(d) Determine the largest subset S of ℂ such that (S, ◊) is a group. [3]

6(a) The surface S has equation $x^2 + y^2 + z^2 = xyz - 1$.

Show that $(2z - xy)\left(x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}\right) = 2\left(1 + z^2\right)$. [6]

(b) Deduce that S has no stationary point. [2]

7 Solve the simultaneous linear congruences $x \equiv 1 \pmod 3$, $x \equiv 5 \pmod{11}$, $2x \equiv 5 \pmod{17}$. [6]

8(a) The points P, Q and R have position vectors $\mathbf{p} = 2\mathbf{i} + \mathbf{j} + 5\mathbf{k}$, $\mathbf{q} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + t\mathbf{k}$ respectively, relative to the origin O.

Determine the value(s) of t

The line OR is parallel to $\mathbf{p} \times \mathbf{q}$. [2]

(b) The volume of tetrahedron $OPQR$ is 13. [4]

9(a) The following Cayley table is for G, a group of order 6. The identity element is e and the group is generated by the elements a and b.

G	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

List all the proper subgroups of G. [4]

(b) State another group of order 6 to which G is isomorphic. [1]

10 In this question you must show detailed reasoning.

Express the number 41723_{10} in hexadecimal (base 16). [3]

11(a)

The sequence $\{u_n\}$ of positive real numbers is defined by $u_1 = 1$ and $u_{n+1} = \dfrac{2u_n + 3}{u_n + 2}$ for $n \geqslant 1$.

Prove by induction that $u_n^2 - 3 < 0$ for all positive integers n. [6]

(b) By considering $u_{n+1} - u_n$, use the result of part (a) to show that $u_{n+1} > u_n$ for all positive integers n. [3]

(c) The sequence $\{u_n\}$ has a limit for $n \to \infty$.

Find the limit of the sequence $\{u_n\}$ as $n \to \infty$. [2]

(d) Describe as fully as possible the behaviour of the sequence $\{u_n\}$. [1]

12(a) Throughout this question, n is a positive integer.

Explain why $n^5 \equiv n \pmod 5$. [1]

(b) By proving that $n^5 \equiv n \pmod{2}$, show that $n^5 \equiv n \pmod{10}$. [3]

(c) (i) Prove that $n^5 - n$ is divisible by 30 for all positive integers n. [5]

(ii) Is there an integer N, greater than 30, such that $n^5 - n$ is divisible by N for all positive integers n?
Justify your answer. [1]

13(a) The group G consists of the set {3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36} under $\times_{39}$, the operation of multiplication modulo 39.

List the possible orders of proper subgroups of G, justifying your answer. [2]

(b) List the elements of the subset of G generated by the element 3. [1]

(c) State the identity element of G. [1]

(d) Determine the order of the element 18. [2]

(e) Find the two elements g_1 and g_2 in G which satisfy $g \times_{39} g = 3$. [3]

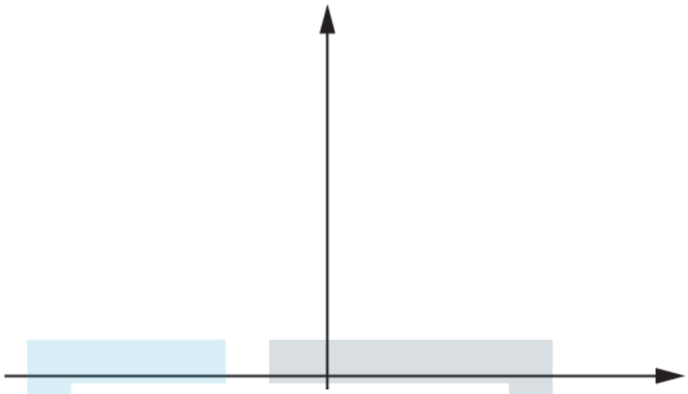
(f) The group H consists of the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12} under $\times_{13}$, the operation of multiplication modulo 13. You are given that G is isomorphic to H.

A student states that G is isomorphic to H because each element $3x$ in G maps directly to the element x in H (for x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12).

Explain why this student is incorrect. [1]

14(a) A designer intends to manufacture a product using a 3-D printer. The product will take the form of a surface S which must meet a number of design specifications. The designer chooses to model S with the equation $z = y \cosh x$ for $-\ln 20 \le x \le \ln 20$, $-2 \le y \le 2$.

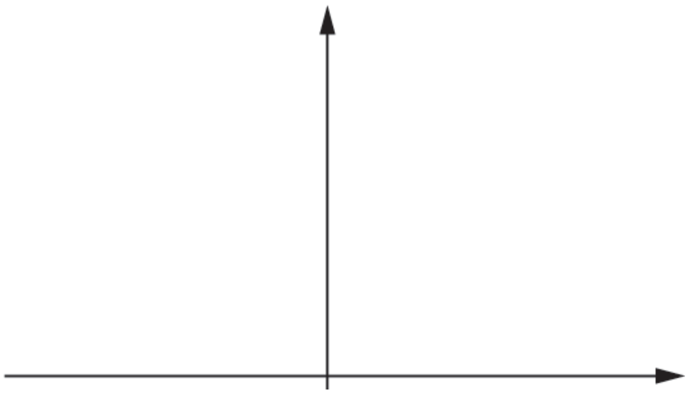
(i) On the axes provided below, sketch the section of S given by $y = 1$. [1]



(ii) One of the design specifications of the product is that this section should have a length no greater than 20 units.

Determine whether the product meets this requirement according to the model. [4]

(b) (i) On the axes provided below, sketch the contour of S given by $z = 1$. [1]



(ii) When this contour is rotated through 2π radians about the x-axis, the surface T is generated. The surface area of T is denoted by A.

Show that A can be written in the form $k\pi \int_0^{\ln 20} \frac{1}{\cosh^3 x}\sqrt{\cosh^4 x + \cosh^2 x - 1}\,\mathrm{d}x$ for some integer k to be determined. [5]

(iii) A second design specification is that the surface area of T must not be greater than 20 square units. Use your calculator to decide whether the product meets this requirement according to the model. [2]

15(a) Points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, relative to origin O.

It is given that $\mathbf{b} \times \mathbf{c} = \mathbf{a}$ and that $|\mathbf{a}| = 3$.

Determine each of the following as either a single vector or a scalar quantity.

(i) $\mathbf{c} \times \mathbf{b}$ [1]

(ii) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ [2]

(iii) $\mathbf{a} \bullet (\mathbf{b} \times \mathbf{c})$ [2]

(b) Describe a geometrical relationship between the points O, A, B and C which can be deduced from

(i) the statement $\mathbf{b} \times \mathbf{c} = \mathbf{a}$, [1]

(ii) the result of (a)(iii). [1]

16(a)

For integers $n \geq 0$, $I_n = \int_0^1 \frac{x^n}{1+x^2} \mathrm{d}x$.

For integers $n \geq 2$, show that $I_n + I_{n-2} = \frac{1}{n-1}$ [3]

(b) (i) Determine the exact value of I_{10}. [4]

(ii) Deduce that $\pi < 3\frac{107}{315}$. [2]

Exam Papers Practice

17(a) For $x, y \in \mathbb{R}$, the function f is given by $f(x, y) = 2x^2 y^7 + 3x^5 y^4 - 5x^8 y$.

Prove that $x\, f_x + y\, f_y = n\, f$, where n is a positive integer to be determined. [5]

(b) Show that $x\, f_{xx} + y\, f_{xy} = (n - 1)\, f_x$. [4]

18(a) The following Cayley table is for a set $\{a, b, c, d\}$ under a suitable binary operation.

	a	b	c	d
a	b		a	
b				
c			c	
d	d			a

Given that the Latin square property holds for this Cayley table, complete the table above. [4]

(b) Using your completed Cayley table, explain why the set does not form a group under the binary operation. [1]

19(a) Prove that $2(p-2)^{p-2} \equiv -1 \pmod{p}$, where p is an odd prime. [4]

(b) Find two odd prime factors of the number $N = 2 \times 34^{34} - 2^{15}$. [7]

20(a)

The points $P\left(\frac{1}{2}, \frac{13}{24}\right)$ and $Q\left(\frac{3}{2}, \frac{31}{24}\right)$ lie on the curve $y = \frac{1}{3}x^3 + \frac{1}{4x}$.

The area of the surface generated when arc PQ is rotated completely about the x-axis is denoted by A.

Find the exact value of A. Give your answer as a rational multiple of π. [4]

(b) Student X finds an approximation to A by modelling the arc PQ as the straight line segment PQ, then rotating this line segment completely about the x-axis to form a surface.

Find the approximation to A obtained by student X. Give your answer as a rational multiple of π. [4]

(c) Student Y finds a second approximation to A by modelling the original curve as the line $y = M$, where M is the mean value of the function $f(x) = \frac{1}{3}x^3 + \frac{1}{4x}$, then rotating this line completely about the x-axis to form a surface.

Find the approximation to A obtained by student Y. Give your answer correct to four decimal places. [4]

21(a)

For the vectors $\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix}$, calculate

- p • q × r ,
- p × (q × r) ,
- (p × q) × r. [6]

(b) State whether the vector product is associative for three-dimensional column vectors with real components. Justify your answer. [1]

(c) It is given that a, b and c are three-dimensional column vectors with real components.

Explain geometrically why the vector a × (b × c) must be expressible in the form λb + μc, where λ and μ are scalar constants.

[2]

(d) It is given that the following relationship holds for a, b and c.

a × (b × c) = (a • c) b − (a • b) c (*)

Find an expression for (a × b) × c in the form of (*). [3]

22(a) The group G consists of a set S together with $\times_{80}$, the operation of multiplication modulo 80. It is given that S is the smallest set which contains the element 11.

By constructing the Cayley table for G, determine all the elements of S. [5]

(b) The Cayley table for a second group, H, also with the operation $\times_{80}$, is shown below.

$\times_{80}$	1	9	31	39
1	1	9	31	39
9	9	1	39	31
31	31	39	1	9
39	39	31	9	1

Use the two Cayley tables to explain why G and H are not isomorphic. [2]

(c) (i) List

- all the proper subgroups of G,
- all the proper subgroups of H. [3]

(ii) Use your answers to (c) (i) to give another reason why G and H are not isomorphic. [1]

23(a) Solve the second-order recurrence relation $T_{n+2} + 2T_n = -87$ given that $T_0 = -27$ and $T_1 = 27$. [8]

(b) Determine the value of T_{20}. [2]

24(a) Solve $7x \equiv 6 \pmod{19}$. [2]

(b) Show that the following simultaneous linear congruences have no solution.

$x \equiv 3 \pmod{4}$, $x \equiv 4 \pmod{6}$. [2]

25(a) A surface has equation $z = \mathrm{f}(x\ y,)$ where $\mathrm{f}(x\ y,\ ) = x^2\sin y + 2y\cos x$.

Determine f_x, f_y, f_{xx}, f_{yy}, f_{xy} and f_{yx}. [5]

(b) (i)

Verify that z has a stationary point at $\left(\frac{1}{2}\pi, \frac{1}{2}\pi, \frac{1}{4}\pi^2\right)$. [3]

(ii) Determine the nature of this stationary point. [3]

26

The sequence $\{u_n\}$, is defined by $u_0 = 2$, $u_1 = 5$ and $u_n = \dfrac{1+u_{n-1}}{u_{n-2}}$ for $n \geq 2$.
Prove that the sequence is periodic with period 5. [4]

27 For each value of t, the surface S_t has equation $z = tx^2 + y^2 + 3xy - y$.

(a) Verify that there are no stationary points on S_t when $t = \frac{9}{4}$. [4]

(b) Determine, as t varies, the nature of any stationary point(s) of S_t.

(You do not have to find the coordinates of the stationary points.) [7]

28

For positive integers n, the integrals I_n are given by $I_n = \int_1^5 x^n \sqrt{2+x^2}\,dx$.

(a) Show that $I_1 = 26\sqrt{3}$. [3]

(b) Prove that, for $n \geq 3$, $(n + 2) I_n = 3\sqrt{3}(27 \times 5^{n-1} - 1) - 2(n-1)I_{n-2}$. [6]

(c) Determine the exact value of I_5 as a rational multiple of $\sqrt{3}$. [4]

29 Torque is a vector quantity that measures how much a force acting on an object causes that object to rotate.

The torque (about the origin), T, exerted on an object is given by $T = p \times F$, where F is the force acting on the object and p is the position vector of the point at which F is applied to the object.

The points A and B, with position vectors $a = 3i+4j+k$ and $b = 3i+5j+k$ are on the surface of a rock. The force $F_1 = 6i+7j-3k$ is applied to the rock at A while the force $F_2 = -7i-10j+2k$ is applied to the rock at B.

(a) Find the torque (about the origin) exerted on the rock by F_1. [2]

(b) Determine which of the two forces F_1 and F_2 exerts a torque (about the origin) of greater magnitude on the rock. [3]

(c) Show that the torque (about the origin) is the same as your answer to part (a) when F_1 acts on the rock at any point on the line $r = a+\lambda p$, where p is a vector in the same direction as F_1. [3]

A third force F_3 is now applied to the rock at A, which exerts zero torque (about the origin).

(d) Show that F_3 must act in the direction of the line through A and the origin. [2]

Exam Papers Practice

30 The set L consists of all points (x, y) in the cartesian plane, with $x \neq 0$. The operation $\diamond$ is defined by $(a, b) \diamond (c, d) = (ac, b + ad)$ for $(a, b), (c, d) \in L$.

(a) (i) Show that L is closed under $\diamond$. [1]

(ii) Prove that $\diamond$ is associative on L. [4]

(iii) Find the identity element of L under $\diamond$. [2]

(iv) Find the inverse element of (a, b) under $\diamond$. [3]

(b) Find a subgroup of $(L, \diamond)$ of order 2. [2]

31 (a) Show that $10^2 \equiv 6 \pmod{47}$. [2]

(b) Determine the integer r, with $0 < r < 47$, such that $6r \equiv 1 \pmod{47}$. [2]

(c) Determine the least positive integer n for which $10^n \equiv 1$ or $-1 \pmod{47}$. [5]

32 A sequence $\{u_n\}$, is given by $u_{n+1} = 4u_n + 1$ for $n \geq 1$ and $u_1 = 3$. (*)

(a) Find the values of u_2, u_3 and u_4. [1]

(b) Solve the recurrence system (*). [5]

(c) (i) Prove by induction that each term of the sequence can be written in the form $(10m + 3)$ where m is an integer. [5]

(ii) Show that no term of the sequence is a square number. [2]

33

A surface has equation $z = x \tan y$ for $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$.

(a) Find

- $\frac{\partial z}{\partial x}$,
- $\frac{\partial z}{\partial y}$. [2]

(b) Find in cartesian form, the equation of the tangent plane to the surface at the point where $x = 1$ and $y = \frac{1}{4}\pi$. [5]

34 The members of the family of the sequences $\{u_n\}$ satisfy the recurrence relation

$$u_{n+1} = 10u_n - u_{n-1} \text{ for } n \geq 1. \qquad (*)$$

(a) Determine the general solution of (*). [3]

(b) The sequences $\{a_n\}$ and $\{b_n\}$ are members of this family of sequences, corresponding to the initial terms $a_0 = 1$, $a_1 = 5$ and $b_0 = 0$, $b_1 = 2$ respectively.

(i) Find the next two terms of each sequence. [1]

(ii) Prove that, for all non-negative integers n, $(a_n)^2 - 6(b_n)^2 = 1$. [8]

(iii) Determine $\lim_{n \to \infty}\left(\frac{a_n}{b_n}\right)$. [2]

35 A class of students is set the task of finding a group of functions, under composition of functions, of order 6. Student P suggests that this can be achieved by finding a function f for which $f^6(x) = x$ and using this as a generator for the group.

(a) Explain why the suggestion by Student P might not work. [2]

Student Q observes that their class has already found a group of order 6 in a previous task; a group consisting of the powers of a particular, non-singular 2×2 real matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, under the operation of matrix multiplication.

(b) Explain why such a group is only possible if $\det(\mathbf{M}) = 1$ or -1. [2]

(c) Write down values of a, b, c and d that would give a suitable matrix M for which $M^6 = I$ and $\det(M) = 1$. [1]

Student Q believes that it is possible to construct a rational function f in the form $f(x) = \frac{ax+b}{cx+d}$ so that the group of functions is isomorphic to the matrix group which is generated by the matrix M of part (c).

(d) (i) Write down and simplify the function f that, according to Student Q, corresponds to M. [1]

(ii) By calculating M^3, show that Student Q's suggestion does not work. [2]

(iii) Find a different function f that will satisfy the requirements of the task. [4]

36

(a) You are given that $N = \binom{p-1}{r}$, where p is a prime number and r is an integer such that $1 \le r \le p - 1$. By considering the number $N \times r!$, prove that $N \equiv (-1)^r \pmod{p}$. [5]

(b) You are given that $M = \binom{2p}{p}$, where p is an odd prime number. Prove that $M \equiv 2 \pmod{p}$. [6]

37 The points A, B, C and P have coordinates $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ and (a, b, c) respectively, where a, b and c are positive constants.

The plane Π contains A, B and C.

(a) (i) Use the scalar triple product to determine

- the volume of tetrahedron $OABC$,
- the volume of tetrahedron $PABC$. [5]

(ii) Hence show that the distance from P to Π is twice the distance from O to Π. [2]

(b) (i) Determine a vector which is normal to Π. [2]

(ii) Hence determine, in terms of a, b and c only, the distance from P to Π. [3]

38 The function $w = \mathrm{f}(x, y, z)$ is given by $\mathrm{f}(x, y, z) = x^2yz + 2xy^2z + 3xyz^2 - 24xyz$, for $x, y, z \neq 0$.

(a) (i) Find

- f_x,
- f_y,
- f_z. [4]

(ii) Hence find the values of a, b, c and d for which w has a stationary value when $d = \mathrm{f}(a, b, c)$. [5]

(b) You are given that this stationary value is a local minimum of w. Find values of x, y and z which show that it is not a global minimum of w. [2]

39 In this question, you must show detailed reasoning.

A curve is defined parametrically by $x = t^3 - 3t + 1$, $y = 3t^2 - 1$, for $0 \le t \le 5$. Find, in exact form,

(a) the length of the curve, [6]

(b) the area of the surface generated when the curve is rotated completely about the x-axis. [4]

40 (a) Write the number 100011_n, where $n \geq 2$, as a polynomial in n. [1]

(b) Show that $n^2 + n + 1$ is a factor of this expression. [2]

(c) Hence show that 100011_n is composite in any number base $n \geq 2$. [2]

41 A survey is carried out into the length of time for which customers wait for a response on a telephone helpline. A statistician who is analysing the results of the survey starts by modelling the waiting time, x minutes, by an exponential distribution with probability density function (PDF)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geqslant 0, \\ 0 & x < 0. \end{cases}$$

(a) In this question you must show detailed reasoning.

The mean waiting time is found to be 5.0 minutes. Show that $\lambda = 0.2$. [4]

(b) Use the model to calculate the probability that a customer has to wait longer than 20 minutes for a response. [2]

In practice it is found that no customer waits for more than 15 minutes for a response. The statistician constructs an improved model to incorporate this fact.

(c) On the diagram below, sketch the following, labelling the curves clearly:

(i) the PDF of the model using the exponential distribution, [1]

(ii) a possible PDF for the improved model. [2]

42

The set M contains all matrices of the form X^n, where $\mathbf{X} = \frac{1}{\sqrt{3}}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$ and n is a positive integer.

(a) Show that M contains exactly 12 elements. [4]

(b) Deduce that M, together with the operation of matrix multiplication, form a cyclic group G. [5]

(c) Determine all the proper subgroups of G. [5]

43 In this question you must show detailed reasoning.

It is given that $I_n = \int_0^{\sqrt{3}} t^n \sqrt{1+t^2}\, dt$ for integers $n \geq 0$.

(a) Show that $I_1 = \frac{7}{3}$. [2]

(b) Prove that, for $n \geq 2$, $(n+2)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$. [6]

The curve C is defined parametrically by

$$x = 10t^3, \quad y = 15t^2 \quad \text{for } 0 \leqslant t \leqslant \sqrt{3}.$$

When the curve C is rotated through 2π radians about the x-axis, a surface of revolution is formed with surface area A.

(c) Determine

- the values of the integers k and m such that $A = k\pi\, I_m$,
- the exact value of A. [6]

44 (a) (i) Solve the recurrence relation

$$X_{n+2} = 1.3X_{n+1} + 0.3X_n \text{ for } n \geqslant 0$$

given that $X_0 = 12$ and $X_1 = 1$. [6]

(ii) Show that the sequence $\{X_n\}$ approaches a geometric sequence as n increases. [2]

The recurrence relation in part (a) models the projected annual profit for an investment company, so that X_n represents the profit (in £) at the end of year n.

(b) (i) Determine the number of years taken for the projected profit to exceed one million pounds. [2]

(ii) Compare your answer to part (b)(i) with the corresponding figure given by the geometric sequence of part (a)(ii). [2]

(c) (i) In a modified model, any non-integer values obtained are rounded down to the nearest integer at each step of the process. Write down the recurrence relation for this model. [1]

(ii) Write down the recurrence relation for the model in which any non-integer values obtained are rounded up to the nearest integer at each step of the process. [1]

(iii) Describe a situation that might arise in the implementation of part (c)(ii) that would result in an incorrect value for the next X_n in the process. [1]

45 (a) (i) Find all the quadratic residues modulo 11. [2]

(ii) Prove that the equation $y^5 = x^2 + 2017$ has no solution in integers x and y. [5]

(b) In this question you must show detailed reasoning.

The numbers M and N are given by

$$M = 11^{12} - 1 \text{ and } N = 3^2 \times 5 \times 7 \times 13 \times 61.$$

Prove that M is divisible by N. [5]

46

The surface S has equation $z = \frac{x}{y}\sin y + \frac{y}{x}\cos x$ where $0 < x \le \pi$ and $0 < y \le \pi$.

(a) Find

- $\frac{\partial z}{\partial x}$,

- $\frac{\partial z}{\partial y}$. [4]

(b) Determine the equation of the tangent plane to S at the point A where $x = y = \frac{1}{4}\pi$. Give your answer in the form $ax + by + cz = d$ where a, b, c and d are exact constants. [5]

(c) Write down a normal vector to S at A. [1]

47 Four points A, B, C and D have coordinates $(1, 2, 5)$, $(3, 4, -4)$, $(6, 2, 3)$ and $(0, 3, 7)$ respectively. Find the volume of tetrahedron $ABCD$. [5]

48 Determine the solution of the simultaneous linear congruences

$$x \equiv 4 \pmod{7}, x \equiv 25 \pmod{41}.$$ [5]

49

It is given that $I_n = \int_0^1 x^n \sqrt{1-x}\,\mathrm{d}x$ for $n \geqslant 0$.

(i) Show that $I_n = \dfrac{2n}{2n+3} I_{n-1}$. [5]

(ii) Deduce that $I_n < I_{n-1}$. [2]

(iii) Show that $I_4 = \frac{256}{3465}$. [3]

50 A group G has the elements $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ where $a, b \in \{1, -1, \mathrm{i}, -\mathrm{i}\}$. The group operation is matrix multiplication.

The subset H consists of the matrices with $a = 1$.

(i) State the order of G. [1]

(ii) Show that H is a subgroup of G. [3]

K is a proper subgroup of G such that H is a proper subgroup of K.

(iii) Show that K must have order 8. [4]

(iv) Show that there is only one such subgroup K and identify its elements. [6]

51 The group G consists of the set $\{1, 5, 7, 11\}$ combined under multiplication modulo 12.

(i) Draw the group table for G. [2]

The group H consists of the set $\{1, 3, 5, 7\}$ combined under multiplication modulo 8.

(ii) Determine whether G and H are isomorphic. [3]

52 A psychologist investigated the scores of pairs of twins on an aptitude test.
Seven pairs of twins were chosen randomly, and the scores are given in the following table.

Elder twin	65	37	60	79	39	40	88
Younger twin	58	39	61	62	50	26	84

(a) Carry out an appropriate Wilcoxon test at the 10% significance level to investigate whether there is evidence of a difference in test scores between the elder and the younger of a pair of twins. [6]

(b) Explain the advantage in this case of a Wilcoxon test over a sign test. [1]

53 (a) (i) Prove that $p \equiv \pm 1 \pmod 6$ for all primes $p > 3$. [2]

(ii) Hence or otherwise prove that $p^2 - 1 \equiv 0 \pmod{24}$ for all primes $p > 3$. [3]

(b) Given that p is an odd prime, determine the residue of 2^{p^2-1} modulo p. [4]

(c) Let p and q be distinct primes greater than 3. Prove that $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$. [5]

54

The set X consists of all 2×2 matrices of the form $\begin{pmatrix} x & -y \\ y & x \end{pmatrix}$, where x and y are real numbers which are not both zero.

(a) (i) The matrices $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ and $\begin{pmatrix} c & -d \\ d & c \end{pmatrix}$ are both elements of X.

Show that $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}$ for some real numbers p and q to be found in terms of a, b, c and d. [2]

(ii) Prove by contradiction that p and q are not both zero. [5]

(b) Prove that X, under matrix multiplication, forms a group G.
[You may use the result that matrix multiplication is associative.] [4]

(c) Determine a subgroup of G of order 17. [2]

55 In order to rescue them from extinction, a particular species of ground-nesting birds is introduced into a nature reserve. The number of breeding pairs of these birds in the nature reserve, t years after their introduction, is an integer denoted by N_t. The initial number of breeding pairs is given by N_0.

An initial discrete population model is proposed for N_t.

Model I: $N_{t+1} = \frac{6}{5} N_t \left(1 - \frac{1}{900} N_t\right)$

(a) (i) For Model I, show that the steady state values of the number of breeding pairs are 0 and 150. [3]

(ii) Show that $N_{t+1} - N_t < 150 - N_t$ when N_t lies between 0 and 150. [3]

(iii) Hence find the long-term behaviour of the number of breeding pairs of this species of birds in the nature reserve predicted by Model I when $N_0 \in (0, 150)$. [2]

An alternative discrete population model is proposed for N_t.

Model II: $N_{t+1} = \mathrm{INT}\left(\frac{6}{5} N_t \left(1 - \frac{1}{900} N_t\right)\right)$

(b) (i) Given that $N_0 = 8$, find the value of N_4 for each of the two models. [2]

(ii) Which of the two models gives values for N_t with the more appropriate level of precision? [1]

56 A surface S has equation $z = \mathrm{f}(x, y)$, where $\mathrm{f}(x, y) = 2x^2 - y^2 + 3xy + 17y$. It is given that S has a single stationary point, P.

(a) (i) Determine the coordinates of P. [5]

(ii) Determine the nature of P. [3]

(b) Find the equation of the tangent plane to S at the point $Q(1, 2, 38)$. [2]

57 In this question you must show detailed reasoning.

It is given that $I_n = \int_0^{\pi} \sin^n \theta \, d\theta$ for $n \geq 0$.

(a) Prove that $I_n = \dfrac{n-1}{n} I_{n-2}$ for $n \geq 2$. [5]

(b) (i) Evaluate I_1. [2]

(ii) Use the reduction formula to determine the exact value of $\int_0^{\pi} \cos^2\theta \sin^5\theta \, d\theta$. [2]

58 (a) Solve the recurrence relation $u_{n+2} = 4u_{n+1} - 4u_n$ for $n \geq 0$, given that $u_0 = 1$ and $u_1 = 1$. [4]

(b) Show that each term of the sequence $\{u_n\}$ is an integer. [2]

59

Given $z = x \sin y + y \cos x$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} + z = 0$. [5]

60 Find the volume of tetrahedron OABC, where O is the origin, A = (2, 3, 1), B = (−4, 2, 5) and C = (1, 4, 4) [3]

61 A curve is given by $x = t^2 - 2\ln t$, $y = 4t$ for $t > 0$. When the arc of the curve between the points where $t = 1$ and $t = 4$ is rotated through 2π radians about the x-axis, a surface of revolution is formed with surface area A. Given that $A = k\pi$, where k is an integer,

- write down an integral which gives A and
- find the value of k. [4]

62 The sequence $u_1, u_2, u_3, \ldots$ is defined by $u_n = 5^n + 2^{n-1}$.

(i) Find u_1, u_2 and u_3.

[2]

(ii) Hence suggest a positive integer, other than 1, which divides exactly into every term of the sequence.

[1]

(iii) By considering $u_{n+1} + u_n$, prove by induction that your suggestion in part (ii) is correct.

[5]

63

The sequence $u_1, u_2, u_3, \ldots$ is defined by $u_1 = 2$ and $u_{n+1} = \frac{u_n}{1 + u_n}$ for $n \geqslant 1$.

(i) Find u_2 and u_3, and show that $u_4 = \frac{2}{7}$.

[3]

(ii) Hence suggest an expression for u_n.

[2]

(iii) Use induction to prove that your answer to part (ii) is correct.

[5]

64 Let A be the set of non-zero integers.

(i) Show that A does not form a group under multiplication.

[2]

(ii) State the largest subset of A which forms a group under multiplication. Show that this is a group.

[3]

65 A non-commutative multiplicative group G of order eight has the elements

$$\{e,\ a,\ a^2,\ a^3,\ b,\ ab,\ a^2b,\ a^3b\},$$

where e is the identity and $a^4 = b^2 = e$.

(i) Show that $ba \neq a^n$ for any integer n.

(ii) Prove, by contradiction, that $ba \neq a^2b$ and also that $ba \neq ab$. Deduce that $ba = a^3b$.

(iii) Prove that $ba^2 = a^2b$.

[2]

[6]

[2]

(iv) Construct group tables for the three subgroups of G of order four.

66 The elements of a group G are polynomials of the form $a + bx + cx^2$, where $a,b,c \in \{0,1,2,3,4\}$. The group operation is addition, where the coefficients are added modulo 5.

(i) State the identity element.

[7]

[1]

(ii) State the inverse of $3 + 2x + x^2$.

[2]

(iii) State the order of G.

[1]

The proper subgroup H contains $2 + x$ and $1 + x$.

(iv) Find the order of H, justifying your answer.

[4]

67 Let G be any multiplicative group. H is a subset of G. H consists of all elements h such that $hg = gh$ for every element g in G.

(i) Prove that H is a subgroup of G.

[8]

Now consider the case where G is given by the following table:

	e	p	q	r	s	t
e	e	p	q	r	s	t
p	p	q	e	s	t	r
q	q	e	p	t	r	s
r	r	t	s	e	q	p
s	s	r	t	p	e	q
t	t	s	r	q	p	e

(ii) Show that H consists of just the identity element.

[4]

68 The group G consists of the set {1, 3, 7, 9, 11, 13, 17, 19} combined under multiplication modulo 20.

(i) Find the inverse of each element.

[3]

(ii) Show that G is not cyclic.

[3]

(iii) Find two isomorphic subgroups of order 4 and state an isomorphism between them.

[5]

69

G consists of the set of matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, where a and b are real and $a^2 + b^2 \neq 0$, combined under the operation of matrix multiplication.

(i) Prove that G is a group. You may assume that matrix multiplication is associative.

[6]

(ii) Determine whether G is commutative.

[2]

(iii) Find the order of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

[3]

70 G consists of the set {1, 3, 5, 7} with the operation of multiplication modulo 8.

(i) Write down the operation table and, assuming associativity, show that G is a group.

[5]

(ii) State the order of each element.

[1]

(iii) Find all the proper subgroups of G.

[1]

The group H consists of the set {1, 3, 7, 9} with the operation of multiplication modulo 10.

(iv) Explaining your reasoning, determine whether H is isomorphic to G.

[2]

71 A commutative group G has order 18. The elements a, b and c have orders 2, 3 and 9 respectively.

(i) Prove that ab has order 6.

[4]

(ii) Show that G is cyclic.

[3]

72 It is given that $I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \, dx$ where n is a positive integer.

(i) By writing $\sec^n x = \sec^{n-2} x \sec^2 x$, or otherwise, show that

[5]

(ii) Show that $I_8 = \frac{96}{35}$.

[3]

(iii) Prove by induction that I_{2n} is rational for all values of $n < 1$.

[4]

73

It is given that $I_n = \int_0^1 x^n e^{-x} dx$ for $n \geqslant 0$.

(i) Show that $I_n = nI_{n-1} + k$ for $n \geqslant 1$, where k is a constant to be determined.

[3]

(ii) Find the exact value of I_3.

[3]

(iii) Find the exact value of $990I_8 - I_{11}$.

[3]

74 You are given that $I_n = \int_0^1 x^n e^{2x}\,dx$ for $n \geqslant 0$.

(i) Show that $I_n = \frac{1}{2}e^2 - \frac{1}{2}nI_{n-1}$ for $n \geqslant 1$.

[4]

(ii) Find I_3 in terms of e.

[4]

75

It is given that, for non-negative integers n, $I_n = \int_0^{\frac{1}{2}\pi} \sin^n x \, dx$.

(i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geqslant 2$.

[3]

(ii) Explain why $I_{2n+1} < I_{2n-1}$.

[2]

(iii) It is given that $I_{2n+1} < I_{2n} < I_{2n-1}$. Take $n = 5$ to find an interval within which the value of π lies.

[6]

76

It is given that $I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \, dx$ for $n \geqslant 0$.

(i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geqslant 2$.

[5]

(ii) Hence find I_{11} as an exact fraction.

[3]

77 The elements of a group G are the complex numbers $a + b\text{i}$ where $a,b \in \{0,1,2,3,4\}$. These elements are combined under the operation of addition modulo 5.

(i) State the identity element and the order of G

[2]

(ii) Write down the inverse of $2 + 4\text{i}$.

[1]

(iii) Show that every non-zero element of G has order 5.

[3]

78 A multiplicative group H has the elements $\{e, a, a^2, a^3, w, aw, a^2w, a^3 w\}$ where e is the identity, elements a and w have orders 4 and 2 respectively and $wa = a^3w$.

(i) Show that $wa^2 = a^2w$ and also that $wa^3 = aw$.

[6]

(ii) Hence show that each of aw, a^2w and a^3w has order 2.

[4]

(iii) Find two non-cyclic subgroups of H of order 4, and show that they are not cyclic.

[4]

79 The sequence $u_1, u_2, u_3, \ldots$ is defined by

Prove by induction that $u_n = 2 \times 3^n - 1$.

$u_1 = 5$ and $u_{n+1} = 3u_n + 2$ for $n \geqslant 1$.

[4]

END OF QUESTION PAPER